

1.2 Fourier-Motzkin Elimination: An Affine Sketch

For a direct proof of Theorem 1.2, the idea is the following. We have to see that every intersection of halfspaces ($\mathcal{H}$ -polyhedron) like $P(A, \mathbf{z})$ is a convex-conical combination ($\mathcal{V}$ -polyhedron) like $\text{conv}(V) + \text{cone}(Y)$, and conversely every convex-conical combination is an intersection of halfspaces.

For the forward direction (“every $\mathcal{V}$ -polyhedron is an $\mathcal{H}$ -polyhedron”) we note that every $\mathcal{V}$ -polyhedron

$$\begin{aligned} \text{conv}(V) + \text{cone}(Y) = \\ \{ \mathbf{x} \in \mathbb{R}^d : \exists \mathbf{t} \in \mathbb{R}^n, \mathbf{u} \in \mathbb{R}^{n'} : \mathbf{x} = V\mathbf{t} + Y\mathbf{u}, \mathbf{t} \geq 0, \mathbf{u} \geq 0, \mathbb{1}\mathbf{t} = 1 \} \end{aligned}$$

can be interpreted as the **projection** of a set

$$\{ (\mathbf{x}, \mathbf{t}, \mathbf{u}) \in \mathbb{R}^{d+n+n'} : \mathbf{x} = V\mathbf{t} + Y\mathbf{u}, \mathbf{t} \geq 0, \mathbf{u} \geq 0, \mathbb{1}\mathbf{t} = 1 \}$$

that quite clearly is an $\mathcal{H}$ -polyhedron. Thus it remains to show that

(I) any projection of an $\mathcal{H}$ -polyhedron is again an $\mathcal{H}$ -polyhedron.

This can be done by the *Fourier-Motzkin elimination* method: projecting down one dimension at a time. We will discuss only the case of projections along coordinate axes, which we need here; the general case can be reduced to this by an affine coordinate transformation.

In this section we give a geometric sketch for the case of affine polyhedra: the nice thing about it (as compared to the — more elegant — version for cones) is that its idea and most of its complications can already be illustrated in dimension 2, so we can provide pictures, and we will. However, instead of doing formulas for this version we switch to the version for cones — and do the proofs there.

We start with an $\mathcal{H}$ -polyhedron $P = P(A, \mathbf{z}) \subseteq \mathbb{R}^d$ and assume that we want to project to $\{ \mathbf{x} \in \mathbb{R}^d : x_k = 0 \} \equiv \mathbb{R}^{d-1}$ along the x_k -axis. The *projection* of $P \subseteq \mathbb{R}^d$ can be defined in great generality; we will only use the cases of coordinate directions, where we use the notation

$$\begin{aligned} \text{proj}_k(P) &:= \{ \mathbf{x} - x_k \mathbf{e}_k : \mathbf{x} \in P \} \\ &= \{ \mathbf{x} \in \mathbb{R}^d : x_k = 0, \exists y \in \mathbb{R} : \mathbf{x} + y \mathbf{e}_k \in P \}. \end{aligned}$$

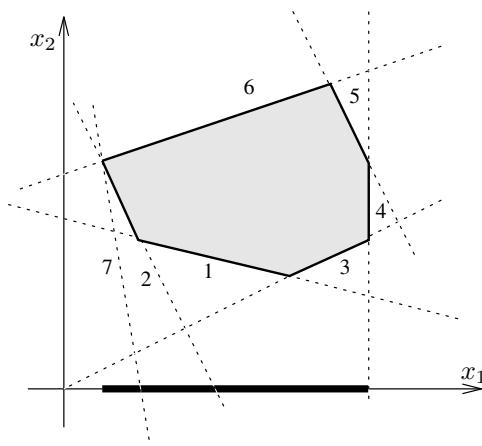
for the *projection* of P in the direction of $\mathbf{e}_k$. The set $\text{proj}_k(P)$ is contained in the hyperplane $H_k = \{ \mathbf{x} \in \mathbb{R}^d : x_k = 0 \}$. A closely related set is the *elimination*

$$\begin{aligned} \text{elim}_k(P) &:= \{ \mathbf{x} - t \mathbf{e}_k : \mathbf{x} \in P, t \in \mathbb{R} \} \\ &= \{ \mathbf{x} \in \mathbb{R}^d : \exists y \in \mathbb{R} : \mathbf{x} + y \mathbf{e}_k \in P \}. \end{aligned}$$

Thus $\text{elim}_k(P)$ is the set of *all* points in $\mathbb{R}^d$ which project to $\text{proj}_k(P)$. In particular, we get an isomorphism $\text{elim}_k(P) \cong \text{proj}_k(P) \times \mathbb{R}$.

For an example, we use the following system of inequalities

$$\begin{array}{rclcl}
 (1) & - & x_1 & - & 4x_2 & \leq & -9 \\
 (2) & - & 2x_1 & - & x_2 & \leq & -4 \\
 (3) & + & x_1 & - & 2x_2 & \leq & 0 \\
 (4) & + & x_1 & & & \leq & 4 \\
 (5) & + & 2x_1 & + & x_2 & \leq & 11 \\
 (6) & - & 2x_1 & + & 6x_2 & \leq & 17 \\
 (7) & - & 6x_1 & - & x_2 & \leq & -6
 \end{array}$$

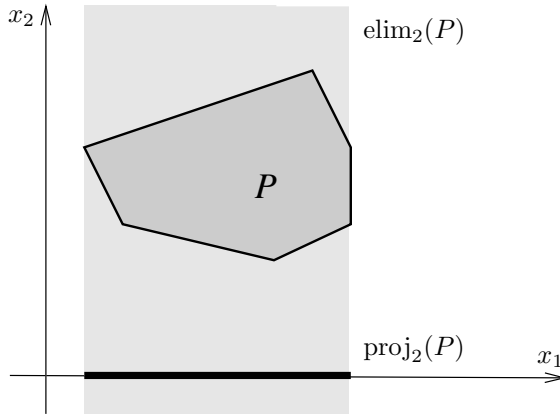


Now assume that we fix some x_1 , and ask for the possible values of x_2 . Then we see that inequality (4) requires $x_1 \leq 4$. All other inequalities can be rewritten to give either an upper bound on x_2 (if the coefficient of x_2 is positive), or lower bound (if the coefficient of x_2 is negative). Furthermore, there is a solution for x_2 if and only if every upper bound for x_2 derived this way is larger than every lower bound.

The sketch on the next page shows the projection of the 2-polytope P to $\text{proj}_2(P)$, by eliminating the x_2 -variable. Here $\text{elim}_2(P)$ is the infinite strip (shaded) of all points that lie above or below $\text{proj}_2(P)$.

Observe how the points of P on any vertical line (i.e., with fixed x_1) are bounded from above and below by inequalities with positive, respectively negative, coefficient a_{i2} . If there is no solution, then some upper bound is smaller than some lower bound: that is, the combination of two inequalities, one with positive and one with negative a_{i2} , leads to a restriction for the possible values of x_1 .

Also note that there is one redundant inequality in the original system: this leads to the effect that the same lower bound on x_1 arises from several different pairs of inequalities.



It is easy to formalize this 2-dimensional description, and to generalize it to arbitrary dimensions. The algebraic treatment rests on the following fact. Consider the coefficients of x_k in our system of inequalities, and assume that $a_{ik} > 0$ and $a_{jk} < 0$. Then the respective inequalities can be rewritten as

$$\mathbf{a}_i \mathbf{x} \leq z_i \quad \longrightarrow \quad a_{ik} x_k \leq a_{ik} x_k - \mathbf{a}_i \mathbf{x} + z_i$$

and

$$\mathbf{a}_j \mathbf{x} \leq z_j \quad \longrightarrow \quad (-a_{jk}) x_k \geq -a_{jk} x_k + \mathbf{a}_j \mathbf{x} - z_j.$$

Here the right-hand sides of the rewritten forms do not depend on x_k , so the first one yields an upper bound on x_k , the second one a lower bound. The combination of the two inequalities (multiplied by the positive coefficients $-a_{jk}$ respectively a_{ik}) yields the condition

$$a_{ik} \mathbf{a}_j \mathbf{x} - a_{ik} z_j \leq -(-a_{jk}) \mathbf{a}_i \mathbf{x} + (-a_{jk}) z_i$$

for “lower bound on x_k below the upper bound,” which is equivalent to the “eliminated inequality”

$$(\mathbf{a}_{ik} \mathbf{a}_j + (-a_{jk}) \mathbf{a}_i) \mathbf{x} \leq a_{ik} z_j + (-a_{jk}) z_i.$$

This is a linear restriction on $\text{elim}_k(P)$. You can see that it is valid without computation: it is a positive combination of two valid inequalities of the original system. However, it is also important to see the geometric content from which it arises.

Now if $\mathbf{x}$ satisfies all these eliminated inequalities, and also those inequalities of the original system which do not involve x_k , then we can conversely find an x_k -coordinate that satisfies $A\mathbf{x} \leq \mathbf{z}$, that is, we have found a complete description of the elimination $\text{elim}_k(P)$ by linear inequalities, and thus proved the following theorem.

Theorem 1.4 (Fourier-Motzkin elimination).

Let $P = P(A, \mathbf{z}) \subseteq \mathbb{R}^d$ be a polyhedron, with $A \in \mathbb{R}^{m \times d}$ and $\mathbf{z} \in \mathbb{R}^m$, and choose $k \leq d$.

Construct the matrix $A^{/k} \in \mathbb{R}^{m' \times d}$ whose rows are

- the rows $\mathbf{a}_i$ of A , for all i with $a_{ik} = 0$, and
- the sums $a_{ik}\mathbf{a}_j + (-a_{jk})\mathbf{a}_i$ for all i, j with $a_{ik} > 0$ and $a_{jk} < 0$,

and let $\mathbf{z}^{/k} \in \mathbb{R}^{m'}$ be the corresponding column vector with entries

- z_i , for all i with $a_{ik} = 0$, and
- $a_{ik}z_j + (-a_{jk})z_i$ for all i, j with $a_{ik} > 0$ and $a_{jk} < 0$.

Then $\text{elim}_k(P) = P(A^{/k}, \mathbf{z}^{/k})$ and

$$\text{proj}_k(P) = P(A^{/k}, \mathbf{z}^{/k}) \cap \{\mathbf{x} \in \mathbb{R}^k : x_k = 0\}.$$

In particular, this says that for $P = P(A, \mathbf{z})$, the projection $\text{proj}_k(P)$ is again an $\mathcal{H}$ -polyhedron. Iterating this, we obtain the forward direction of Theorem 1.2.

The problem with this is that the formulas are messy, partly because we have to deal with the right-hand sides z_i separately. Those will miraculously disappear in the homogeneous version; see the next section.

For the backward direction (“every $\mathcal{H}$ -polyhedron is a $\mathcal{V}$ -polyhedron”), we observe that every $\mathcal{H}$ -polyhedron

$$P(A, \mathbf{z}) = \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} \leq \mathbf{z}\}$$

can be written as the **intersection** of a polyhedron (in fact, a cone)

$$C_0(A) := \left\{ \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} \in \mathbb{R}^{d+m} : A\mathbf{x} \leq \mathbf{w} \right\}$$

with an affine subspace

$$\left\{ \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} \in \mathbb{R}^{d+m} : \mathbf{w} = \mathbf{z} \right\}.$$

The cone $C_0(A)$ is easily seen to be a $\mathcal{V}$ -polyhedron: it can be written as

$$C_0(A) = \text{cone}\left(\left\{\pm \begin{pmatrix} \mathbf{e}_i \\ A\mathbf{e}_i \end{pmatrix} : 1 \leq i \leq d\right\} \cup \left\{\begin{pmatrix} \mathbf{0} \\ \mathbf{e}_j \end{pmatrix} : 1 \leq j \leq m\right\}\right)$$

by decomposing

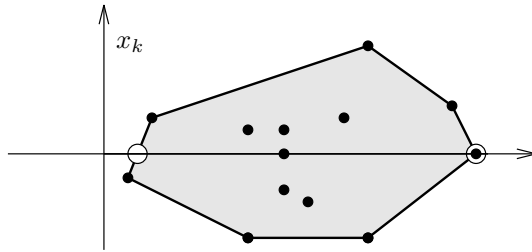
$$\begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} = \sum_{i=1}^d |x_i|(\text{sign}(x_i)) \begin{pmatrix} \mathbf{e}_i \\ A\mathbf{e}_i \end{pmatrix} + \sum_{j=1}^m (w_j - (A\mathbf{x})_j) \begin{pmatrix} \mathbf{0} \\ \mathbf{e}_j \end{pmatrix}.$$

It remains to show that

- (II) any intersection of a $\mathcal{V}$ -polyhedron with an affine subspace is a $\mathcal{V}$ -polyhedron.

The method to prove this, sometimes called the *double description method*, is dual to the method of Fourier-Motzkin elimination (see Motzkin, Raiffa, Thompson & Thrall [416] and Dantzig & Eaves [175]). We restrict our discussion to a very special case, namely the intersection of a polytope $\text{conv}(V)$ with a coordinate hyperplane $H_k = \{\mathbf{x} \in \mathbb{R}^d : x_k = 0\}$, which can then be iterated.

So, we are given a $\mathcal{V}$ -polytope $P = \text{conv}(V) \subseteq \mathbb{R}^d$, and we want to see that $P \cap H_k$ is of the same form. In our sketch the black dots denote the set V , and the bigger white dots denote the two vertices of the intersection of $\text{conv}(V)$ with the hyperplane H_k .



From this we get a little geometric intuition, which suggests that we can write down

$$P \cap H_k = \text{conv}(V^{/k}),$$

where $V^{/k}$ is the matrix (set) of column vectors constructed as

$$V^{/k} := \{\mathbf{v}_i : v_{ki} = 0\} \cup \left\{ \frac{v_{ki}\mathbf{v}_j + (-v_{kj})\mathbf{v}_i}{v_{ki} - v_{kj}} : v_{ki} > 0, v_{kj} < 0 \right\}.$$

For this it is quite clear that we get $P \cap H_k \supseteq \text{conv}(V^{/k})$, but for the converse we have to work a little. We omit this ugly little computation here since you'll see it in the next section: it comes out a little nicer in the homogeneous form (Lemma 1.6). Anyway, this way “in principle” one can give an *explicit* representation of $\mathbf{x}$ as a convex combination of vectors in $V^{/k}$.

This finishes the proof of Theorems 1.1–1.3. $\square$

One can do a similar argument for $P = \text{conv}(V) + \text{cone}(Y)$. However, the corresponding computations become extremely tedious — they are too ugly even to leave them as an exercise. This is why we homogenize and switch to cones, where all difficulties disappear.

1.3 Fourier-Motzkin Elimination for Cones

The main objective of this section is to prove Theorem 1.3. This again has two parts.

For the “forward direction” of Theorem 1.3, let $C = \text{cone}(Y) \subseteq \mathbb{R}^d$ be a $\mathcal{V}$ -cone. We can write it as

$$\begin{aligned} C &= \{Y\mathbf{t} \in \mathbb{R}^d : \mathbf{t} \geq \mathbf{0}\} \\ &= \{\mathbf{x} \in \mathbb{R}^d : \exists \mathbf{t} \in \mathbb{R}^n : \mathbf{t} \geq \mathbf{0}, \mathbf{x} = Y\mathbf{t}\}. \end{aligned}$$

The set $\{(\mathbf{x}, \mathbf{t}) \in \mathbb{R}^{d+n} : \mathbf{t} \geq \mathbf{0}, \mathbf{x} = Y\mathbf{t}\}$ clearly is an $\mathcal{H}$ -cone. Thus $C = \text{cone}(Y)$ can be written as the projection of this cone to the subspace $\{(\mathbf{x}, \mathbf{t}) \in \mathbb{R}^{d+n} : \mathbf{t} = \mathbf{0}\}$. Again, this projection can be formed successively, by projecting with respect to individual t_k -coordinates one by one. Thus it suffices to prove the following lemma.

Lemma 1.5. *If $C = P(A, \mathbf{0})$ is an $\mathcal{H}$ -cone in $\mathbb{R}^d$, then so is the elimination $\text{elim}_k(C) = \{\mathbf{x} - te_k : \mathbf{x} \in C, t \in \mathbb{R}\}$, and thus also the projection $\text{proj}_k(C) = \text{elim}_k(C) \cap H_k$. Namely, we get $\text{elim}_k(C) = P(A^{/k}, \mathbf{0})$ for*

$$A^{/k} := \{\mathbf{a}_i : a_{ik} = 0\} \cup \{a_{ik}\mathbf{a}_j + (-a_{jk})\mathbf{a}_i : a_{ik} > 0, a_{jk} < 0\}.$$

(Here we interpret A and $A^{/k}$ as sets of row vectors.)

Proof. The row vectors in $A^{/k}$ are positive combinations of row vectors in A , hence the corresponding inequalities are valid for C , and thus we get that $C \subseteq P(A^{/k}, \mathbf{0})$. Furthermore, the row vectors in $A^{/k}$ all have the k th component equal to zero (by construction), i.e., the variable x_k does not appear in the system $A^{/k}\mathbf{x} \leq \mathbf{0}$, which proves that $\text{elim}_k(C) \subseteq P(A^{/k}, \mathbf{0})$.

For the converse, let $\mathbf{x} \in P(A^{/k}, \mathbf{0})$, and let $x_k = 0$ (without loss of generality). We claim that $\mathbf{x} - ye_k \in C$ for suitable y . In fact, plugging $\mathbf{x} - ye_k$ into the system $A\mathbf{x} \leq \mathbf{0}$, we find that y has to satisfy

$$\max_i \left\{ \frac{1}{a_{ik}} \mathbf{a}_i \mathbf{x} : a_{ik} > 0 \right\} \leq y \leq \min_j \left\{ \frac{1}{-a_{jk}} (-\mathbf{a}_j) \mathbf{x} : a_{jk} < 0 \right\}.$$

This can be satisfied. Namely, if $a_{ik} > 0$ and $a_{jk} < 0$, then we know that $\frac{1}{a_{ik}} \mathbf{a}_i \mathbf{x} \leq \frac{1}{-a_{jk}} (-\mathbf{a}_j) \mathbf{x}$, which is equivalent to $(a_{ik}\mathbf{a}_j + (-a_{jk})\mathbf{a}_i) \mathbf{x} \leq 0$, which holds because $\mathbf{x} \in P(A^{/k}, \mathbf{0})$. $\square$

Now we proceed to prove the “backward direction” of Theorem 1.3. For this let $C = P(A, \mathbf{0}) \subseteq \mathbb{R}^d$ be an $\mathcal{H}$ -cone. We can write it as

$$\begin{aligned} C &= \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} \leq \mathbf{0}\} \\ &\cong \left\{ \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} \in \mathbb{R}^{d+m} : A\mathbf{x} \leq \mathbf{w} \right\} \cap \left\{ \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} \in \mathbb{R}^{d+m} : \mathbf{w} = \mathbf{0} \right\}. \end{aligned}$$